

Control Technology

Exercises

1 Tank Level Control System

A liquid tank has cross-sectional area A . The inflow is $q_{in}(t)$, outflow is proportional to level $h(t)$.

$$A \frac{dh(t)}{dt} = q_{in}(t) - \frac{1}{R}h(t)$$

Parameters:

$$A = 5, \quad R = 2$$

- (a) Derive the transfer function $G(s) = H(s)/Q_{in}(s)$.
- (b) Determine system type and stability.
- (c) Calculate the step response:
 - (i) steady-state gain
 - (ii) time constant
- (d) Interpret the frequency response.
- (e) PI control design:

$$C(s) = K_P + \frac{K_I}{s}$$

- (i) Derive the closed-loop characteristic equation.
 - (ii) Determine the steady-state error for a step input.
- (f) Engineering interpretation: Why is PI control almost always used for tank systems?

Solution:

- (a)

Laplace:

$$AsH(s) = Q_{in}(s) - \frac{1}{R}H(s)$$

$$H(s) \left(As + \frac{1}{R} \right) = Q_{in}(s)$$

$$G(s) = \frac{H(s)}{Q_{in}(s)} = \frac{1}{As + \frac{1}{R}}$$

Insert values:

$$G(s) = \frac{1}{5s + 0.5}$$

(b)

First order system $\rightarrow$ stable (pole at -0.1)

(c)

DC gain:

$$G(0) = \frac{1}{0.5} = 2$$

Time constant:

$$\tau = \frac{A}{1/R} = AR = 10$$

(d)

First-order low-pass system

Slope: -20 dB/dec

(e)

PI controller closed-loop:

$$L(s) = \frac{K_P s + K_I}{s(5s + 0.5)}$$

Characteristic equation:

$$s(5s + 0.5) + K_P s + K_I = 0$$

$$5s^2 + (0.5 + K_P)s + K_I = 0$$

Steady-state error:

$$e_{ss} = 0$$

for stable system

(f)

PI is used because:

- process is integrating-like or slow
- steady-state offset unacceptable
- P control alone leaves offset

2 Mass-Spring-Damper System

A mechanical system consists of a mass m , damper c , and spring k .

$$m\ddot{x}(t) + c\dot{x}(t) + kx(t) = F(t)$$

Parameters:

$$m = 2, \quad c = 4, \quad k = 6$$

Assume zero initial conditions.

(a) Derive the transfer function

$$G(s) = \frac{X(s)}{F(s)}.$$

(b) Determine system order and stability.

(c) Step response analysis:

- (i) steady-state displacement
- (ii) settling time estimate

(d) Frequency response interpretation:

- (i) system type (low/high-pass)
- (ii) DC gain
- (iii) qualitative high-frequency behavior

(e) Closed-loop with proportional controller $C(s) = K_P$:

- (i) Derive the closed-loop transfer function.
- (ii) Determine the characteristic equation.
- (iii) Determine the stability condition for K_P .

(f) PI controller:

$$C(s) = K_P + \frac{K_I}{s}$$

- (i) Derive the closed-loop characteristic equation.
- (ii) Determine the steady-state error for a step input.

Solution:

(a)

Laplace transform:

$$ms^2X(s) + csX(s) + kX(s) = F(s)$$

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{ms^2 + cs + k}$$

Insert values:

$$G(s) = \frac{1}{2s^2 + 4s + 6}$$

(b)

Order: 2

Stability: all coefficients positive $\rightarrow$ stable

(c)

Steady-state (step input):

$$G(0) = \frac{1}{6}$$

$$x(\infty) = \frac{1}{6}$$

Settling time:

dominant pole approx $\rightarrow$ order 2 stable system:

$$T_s \approx \frac{4}{\zeta\omega_n}$$

(accept qualitative estimate)

(d)

Type: low-pass system

DC gain:

$$G(0) = \frac{1}{6}$$

High frequency:

$$|G(j\omega)| \sim \frac{1}{\omega^2}$$

→ slope -40 dB/dec

(e)

Closed loop:

$$T(s) = \frac{K_P}{2s^2 + 4s + 6 + K_P}$$

Characteristic equation:

$$2s^2 + 4s + 6 + K_P = 0$$

Stability:

all coefficients positive → $K_P > -6$

(practically $K_P > 0$)

(f)

PI controller:

$$C(s) = \frac{K_P s + K_I}{s}$$

Characteristic equation:

$$s(2s^2 + 4s + 6) + K_P s + K_I = 0$$

$$2s^3 + 4s^2 + (6 + K_P)s + K_I = 0$$

Steady-state error:

PI introduces integrator → for step input:

$$e_{ss} = 0$$

3 DC Motor Control System

A separately excited DC motor is described by the following physical relations.

Electrical subsystem:

$$u(t) = L \frac{di(t)}{dt} + Ri(t) + e(t)$$

Back electromotive force:

$$e(t) = K \omega(t)$$

Mechanical subsystem:

$$J \frac{d\omega(t)}{dt} + b \omega(t) = K i(t)$$

The parameters are:

$$J = 0.02, \quad b = 0.2, \quad L = 0.5, \quad R = 1, \quad K = 0.1$$

Assume zero initial conditions.

(a) **Modeling**

Derive the transfer function

$$G(s) = \frac{\Omega(s)}{U(s)}$$

starting from the physical equations.

(b) **System representation**

Insert numerical values and express the transfer function in polynomial form.

Determine the system order.

Is the system stable? Justify your answer.

(c) **Time-domain behavior**

The system is excited by a unit step input.

- (i) Determine the steady-state value of the angular velocity.
- (ii) Estimate the settling time using a first-order approximation.

(d) **Frequency-domain interpretation**

Consider the transfer function obtained in (b).

- (i) What type of system (low-pass, high-pass, etc.) is this?
- (ii) What is the DC gain?
- (iii) Explain qualitatively how the magnitude behaves for low and high frequencies.

(e) **Proportional control**

The motor is controlled in unity feedback with a proportional controller

$$C(s) = K_P.$$

- (i) Derive the closed-loop transfer function.
- (ii) Determine the characteristic equation.
- (iii) Determine the stability condition for K_P .

(f) **PI control**

Now consider a PI controller:

$$C(s) = K_P + \frac{K_I}{s}.$$

- (i) Derive the closed-loop characteristic equation.
- (ii) Explain the effect of the integral term on steady-state error.
- (iii) Determine the steady-state error for a step input (qualitatively or using final value theorem).

Solution:

(a) **Modeling**

Take the Laplace transform under zero initial conditions.

Electrical equation:

$$U(s) = LsI(s) + RI(s) + E(s)$$

Using

$$E(s) = K\Omega(s)$$

gives

$$U(s) = (Ls + R)I(s) + K\Omega(s)$$

Mechanical equation:

$$Js\Omega(s) + b\Omega(s) = KI(s)$$

Thus

$$I(s) = \frac{Js + b}{K}\Omega(s)$$

Substitute into the electrical equation:

$$U(s) = (Ls + R)\frac{Js + b}{K}\Omega(s) + K\Omega(s)$$

Factor out $\Omega(s)$:

$$U(s) = \left(\frac{(Ls + R)(Js + b)}{K} + K \right) \Omega(s)$$

Therefore,

$$\frac{\Omega(s)}{U(s)} = \frac{K}{(Js + b)(Ls + R) + K^2}$$

Hence,

$$G(s) = \frac{K}{(Js + b)(Ls + R) + K^2}$$

(b) System representation

Expand the denominator:

$$\begin{aligned} & (Js + b)(Ls + R) + K^2 \\ &= JJs^2 + (JR + bL)s + bR + K^2 \end{aligned}$$

Insert values:

$$JL = 0.02 \cdot 0.5 = 0.01$$

$$JR = 0.02 \cdot 1 = 0.02$$

$$bL = 0.2 \cdot 0.5 = 0.10$$

$$bR = 0.2 \cdot 1 = 0.20$$

$$K^2 = 0.01$$

Therefore,

$$G(s) = \frac{0.1}{0.01s^2 + 0.12s + 0.21}$$

The highest power of s is two.

Therefore, the system is a second-order system.

(d) Stability

The denominator polynomial is

$$0.01s^2 + 0.12s + 0.21$$

All coefficients are positive.

For a second-order polynomial

$$a_2s^2 + a_1s + a_0$$

the system is stable (all poles in the left half-plane) if

$$a_2 > 0, \quad a_1 > 0, \quad a_0 > 0$$

Since this condition is fulfilled, the system is stable.

(c) Time-domain behavior

Step input steady-state value:

$$G(0) = \frac{0.1}{0.21} \approx 0.476$$

Thus:

$$\omega(\infty) \approx 0.476$$

Settling time approximation:

Normalize denominator:

$$0.01s^2 + 0.12s + 0.21$$

Dominant time constant approximation:

$$\tau \approx \frac{1}{\text{dominant pole}} \approx \frac{1}{2} \approx 0.5 \text{ to } 1 \text{ s}$$

More robust engineering estimate:

$$T_s \approx \frac{4}{\text{dominant pole}} \approx 2 \text{ to } 4 \text{ s}$$

(acceptable qualitative result expected)

(d) Frequency-domain interpretation

Type of system:

Second-order system with no zeros $\rightarrow$ low-pass behavior.

DC gain:

$$G(0) = \frac{0.1}{0.21} \approx 0.476$$

Qualitative behavior:

- For low frequencies: magnitude approximately constant (DC gain)
- For high frequencies: denominator dominates $\rightarrow$ magnitude decreases
- Asymptotic slope: approximately -40 dB/decade

(e) Proportional control

Controller:

$$C(s) = K_P$$

Closed-loop transfer function:

$$T(s) = \frac{K_P G(s)}{1 + K_P G(s)}$$

Insert plant:

$$T(s) = \frac{K_P \cdot 0.1}{0.01s^2 + 0.12s + 0.21 + 0.1K_P}$$

Characteristic equation:

$$0.01s^2 + 0.12s + 0.21 + 0.1K_P = 0$$

Stability condition:

Second-order system stable if all coefficients positive:

$$0.01 > 0, \quad 0.12 > 0, \quad 0.21 + 0.1K_P > 0$$

Thus:

$$K_P > -2.1$$

For practical control (positive gain):

$$K_P > 0$$

(f) **PI control**

Controller:

$$C(s) = K_P + \frac{K_I}{s} = \frac{K_P s + K_I}{s}$$

Open-loop transfer function:

$$L(s) = \frac{K_P s + K_I}{s} \cdot \frac{0.1}{0.01s^2 + 0.12s + 0.21}$$

(i) Characteristic equation:

$$1 + L(s) = 0$$

Multiply through:

$$s(0.01s^2 + 0.12s + 0.21) + 0.1(K_P s + K_I) = 0$$

Expand:

$$0.01s^3 + 0.12s^2 + 0.21s + 0.1K_P s + 0.1K_I = 0$$

Thus:

$$0.01s^3 + 0.12s^2 + (0.21 + 0.1K_P)s + 0.1K_I = 0$$

(ii) Effect of integral term:

The integral term introduces an additional pole at the origin in the open loop.

Consequences:

- eliminates steady-state error for step inputs
- increases system type by 1
- may reduce stability margin

(iii) Steady-state error (step input):

Using final value theorem:

$$e_{ss} = \lim_{s \rightarrow 0} \frac{1}{1 + L(s)}$$

For PI controller:

$$\lim_{s \rightarrow 0} L(s) = \infty$$

due to $1/s$ term.

Thus:

$$e_{ss} = 0$$

for a stable closed-loop system.